SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1105

COURSE OUTCOME COVERED

CO1: *Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 60%

QUESTIONS: 12 **Total Marks:60**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I P.Mal Reddy (192372015) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

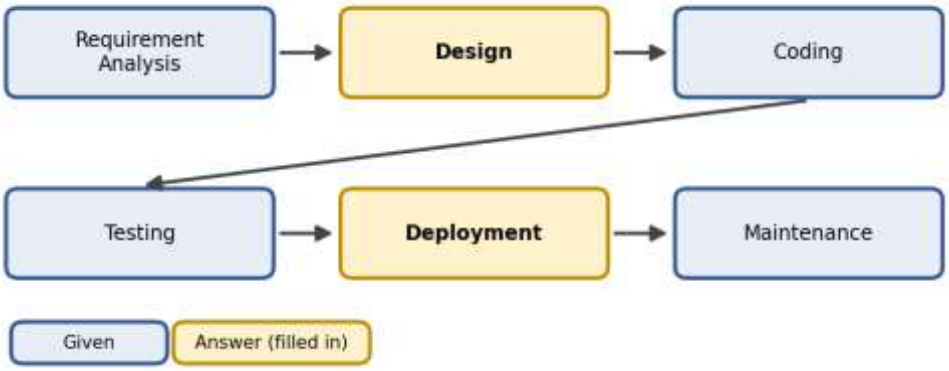
Signature of the student:
P.MAL REDDY

1. Construct a Concept Map

Given:

Requirement Analysis → _____ → Coding → Testing → _____ → Maintenance

Concept Map:



Answer – the missing concepts are:

- Design
- Deployment

Explanation:

- Design converts requirements into system architecture.
- Deployment releases the software to end users.

Analysis:

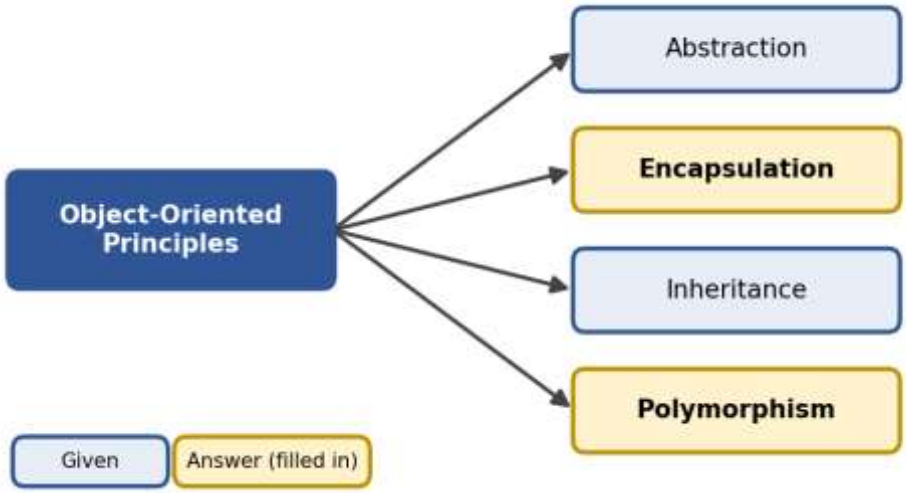
- Ensures systematic software development.
- Reduces project risks.
- Improves software quality and maintainability.

2. Construct a Concept Map

Given:

Object-Oriented Principles → Abstraction → _____ → Inheritance → _____

Concept Map:



Answer – the missing concepts are:

- Encapsulation
- Polymorphism

Explanation:

- Encapsulation protects data from unauthorized access.
- Polymorphism allows one interface to support multiple behaviors.

Analysis:

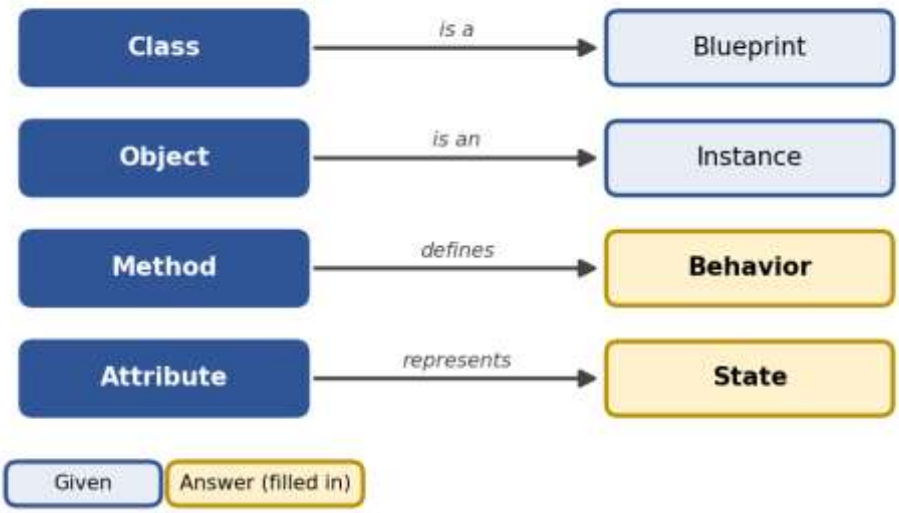
- Improves security and flexibility.
- Enhances code reusability.
- Supports efficient software design.

3. Construct a Concept Map

Given:

Class → Blueprint; Object → Instance; Method → _____; Attribute → _____

Concept Map:



Answer – the missing concepts are:

- Behavior
- State

Explanation:

- Methods define the behavior of objects.
- Attributes represent the state of objects.

Analysis:

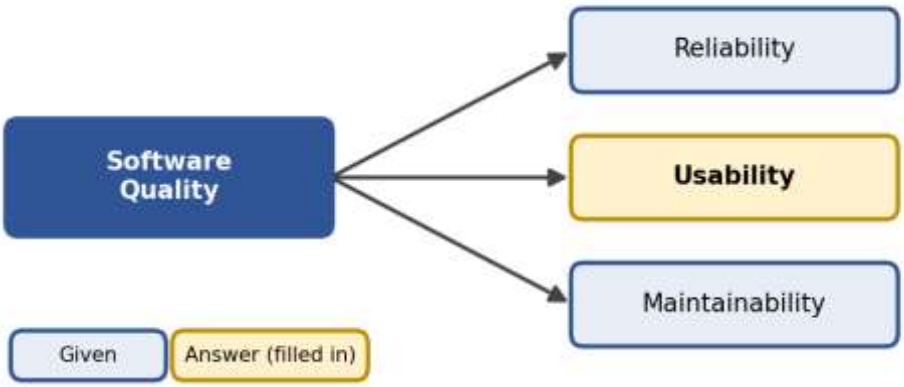
- Helps model real-world entities.
- Defines object characteristics and actions.
- Improves system organization.

4. A Concept Map Shows

Given:

Software Quality → Reliability → _____ → Maintainability

Concept Map:



Answer – the missing concept is:

- Usability

Explanation:

- Usability measures how easily users interact with software.
- Reliable software performs consistently.
- Maintainable software is easy to update.

Analysis:

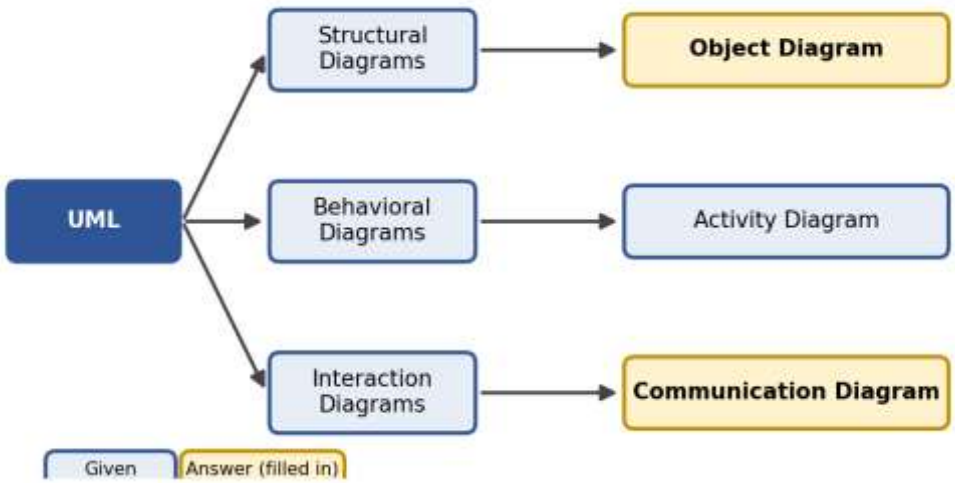
- Improves user satisfaction.
- Enhances product acceptance.
- Contributes to software success.

5. UML Concept Map

Given:

UML → Structural Diagrams → _____; Behavioral Diagrams → Activity Diagram; Interaction Diagrams → _____

Concept Map:



Answer – the missing concepts are:

- Object Diagram
- Communication Diagram

Explanation:

- Object diagrams represent object instances and relationships.
- Communication diagrams show message exchange among objects.

Analysis:

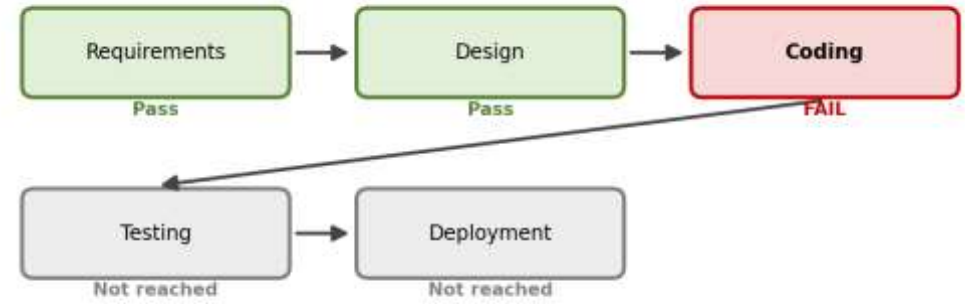
- Improves system visualization.
- Supports object-oriented analysis.
- Facilitates stakeholder communication.

6. Failure Analysis in the Development Process

Given:

Requirements ✓ → Design ✓ → Coding ✗ → Testing ✗ → Deployment ✗

Concept Map:



Explanation:

- Failure occurs during the Coding phase.
- Program logic contains implementation errors.

Analysis:

- Errors affect subsequent phases.
- Testing cannot proceed effectively.
- Software quality decreases.

Solution:

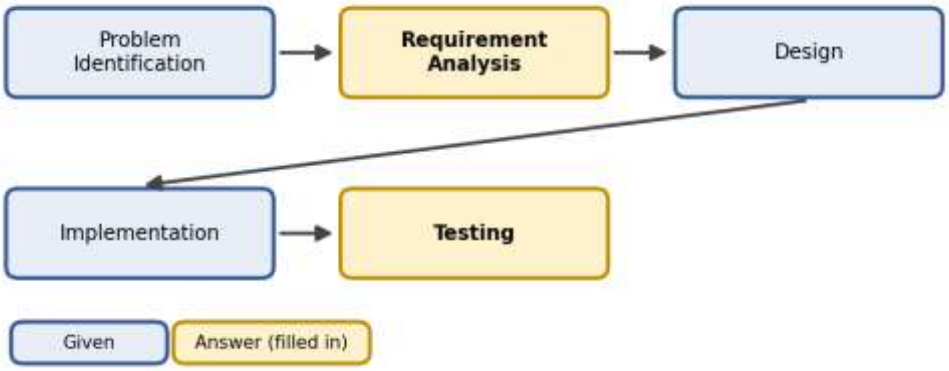
- Review source code.
- Apply coding standards.
- Conduct code inspections before testing.

7. Construct a Concept Map

Given:

Problem Identification → _____ → Design → Implementation → _____

Concept Map:



Answer – the missing concepts are:

- Requirement Analysis
- Testing

Explanation:

- Requirement Analysis identifies user needs and system requirements.
- Testing verifies whether the implemented system meets the specified requirements.

Analysis:

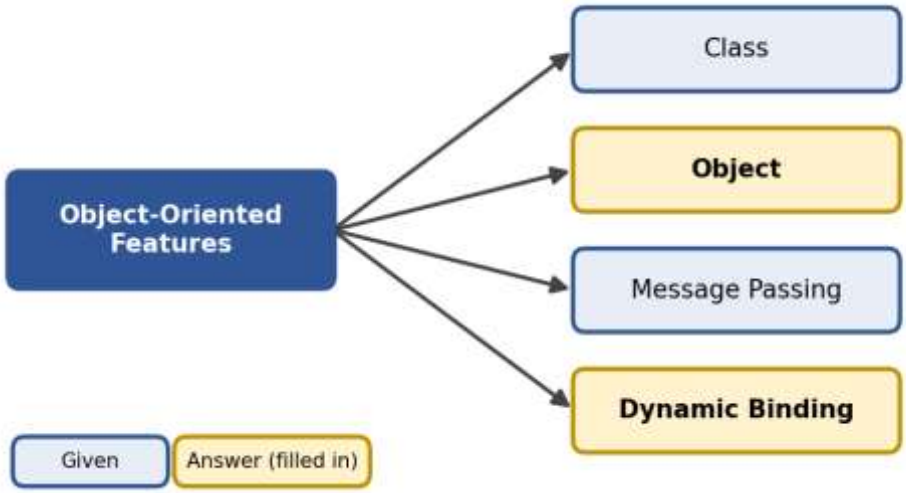
- Improves software quality.
- Reduces development errors.
- Ensures successful project completion.

8. Construct a Concept Map

Given:

Object-Oriented Features → Class → _____ → Message Passing → _____

Concept Map:



Answer – the missing concepts are:

- Object
- Dynamic Binding

Explanation:

- Objects are instances of classes.
- Dynamic binding allows method calls to be resolved at runtime.

Analysis:

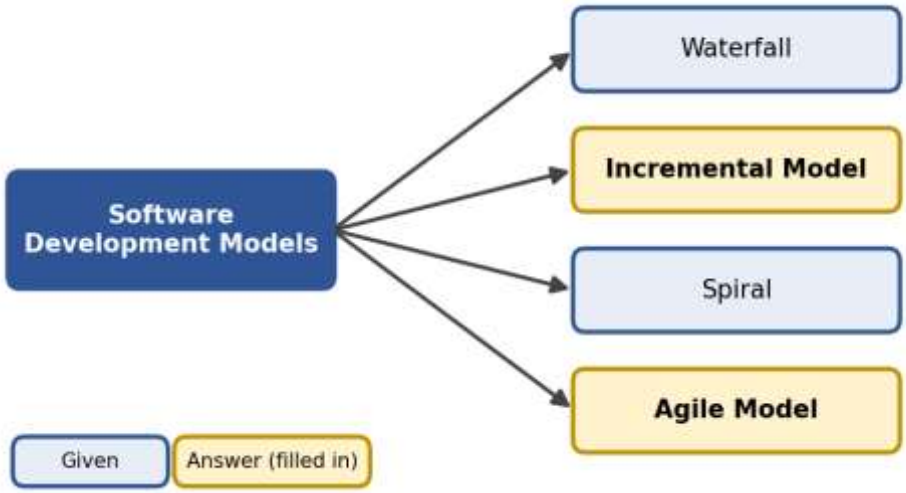
- Supports flexibility in software design.
- Improves code maintainability.
- Enables runtime polymorphism.

9. Construct a Concept Map

Given:

Software Development Models → Waterfall → _____ → Spiral → _____

Concept Map:



Answer – the missing concepts are:

- Incremental Model
- Agile Model

Explanation:

- Incremental development delivers software in stages.
- Agile supports iterative development with continuous customer feedback.

Analysis:

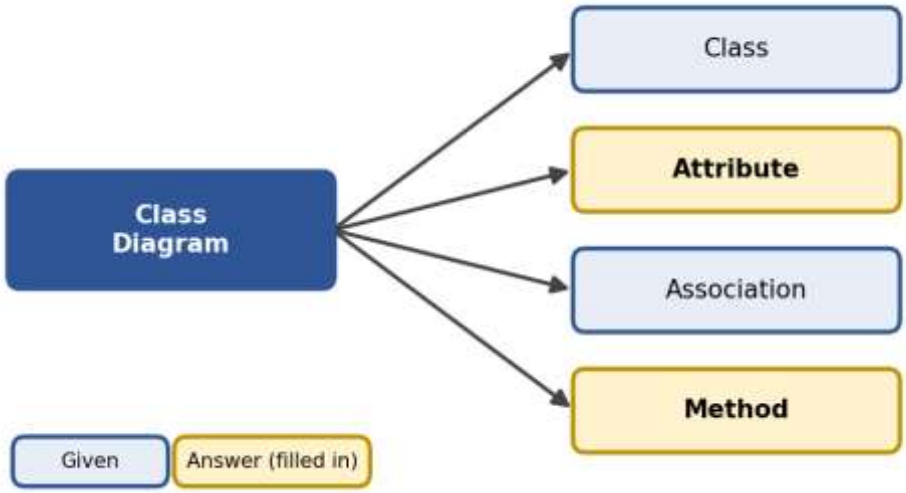
- Enhances adaptability.
- Reduces project risks.
- Improves customer satisfaction.

10. Construct a Concept Map

Given:

Class Diagram → Class → _____ → Association → _____

Concept Map:



Answer – the missing concepts are:

- Attribute
- Method

Explanation:

- Attributes define the properties of a class.
- Methods define the operations performed by a class.

Analysis:

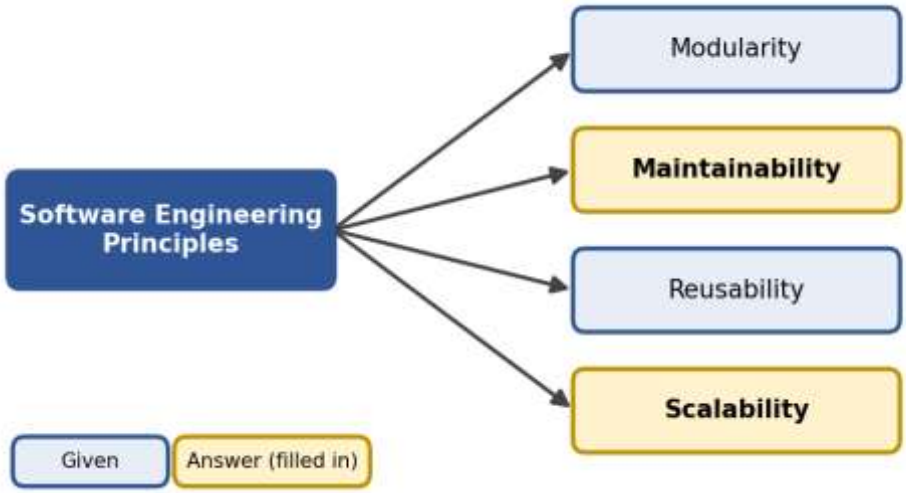
- Represents object structure effectively.
- Improves software documentation.
- Supports object-oriented design.

11. Construct a Concept Map

Given:

Software Engineering Principles → Modularity → _____ → Reusability → _____

Concept Map:



Answer – the missing concepts are:

- Maintainability
- Scalability

Explanation:

- Maintainability simplifies software updates.
- Scalability enables systems to handle increased workload efficiently.

Analysis:

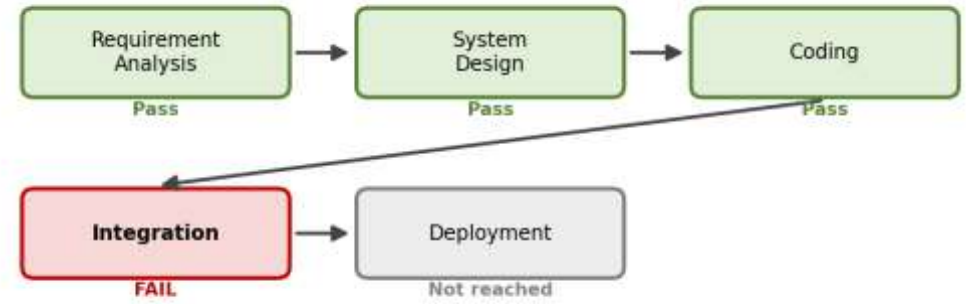
- Improves software performance.
- Reduces maintenance effort.
- Supports long-term software evolution.

12. Failure Analysis in the Development Process

Given:

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✗ → Deployment ✗

Concept Map:



Explanation:

- Failure occurs during the Integration phase.
- Individual modules fail to communicate correctly.

Analysis:

- Causes functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

Solution:

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.